Rikesh Mandal

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Professional Summary

I am a MSc Artificial Intelligence graduate with Distinction and hold a First-Class Honours degree in BSc Computer Networking & IT Security. My academic and professional background combines software engineering, 3D animation, and AI, giving me a broad and practical understanding of CGI, how digital systems are built and how intelligent systems can learn from data and solve complex problems. I started my professional career as a 3D Animator and later pivoted to software development. More recently, my focus has shifted towards machine learning, deep learning, AI and data science. I have designed and implemented systems based on machine learning, deep learning, and software engineering as part of my academic and independent work.

Education

MSc Artificial Intelligence — Distinction (82% or 3.28 / 4.0 GPA), University of East London 2024 – 2025
Dissertation: "An AI-Driven Framework for Automated 3D Model Generation from 2D Images"

BSc Computer Networking & IT Security — First Class (81.94% or 3.28 / 4.0 GPA), London Metropolitan University 2021 – 2023
Final Year Project: "Sign Language to Speech Glove"

Key Projects

Microblog — Full-Stack Flask Web Application | GitHub | Live Demo | API Documentation

Developed and deployed a Flask web application with REST API design, containerized infrastructure, OpenAPI documentation, full-text search, multilingual translation and cloud deployment.

- Designed and implemented REST API endpoints for users, posts, messages, followers/following and token management, using HTTP Basic Authentication for token issuance, Bearer token authentication for protected routes, JSON request/response handling, pagination, and HTTP status codes.
- Modelled relational data with SQLAlchemy ORM and MySQL, implementing foreign-key relationships, migrations with Flask-Migrate/Alembic, database indexing, query optimisation and reusable paginated API responses.
- Containerised the application with Docker and Docker Compose, orchestrating Flask/Gunicorn, MySQL, Elasticsearch and LibreTranslate services with internal service discovery, environment-based configuration and persistent named volumes for database, search-index and translation model data.
- Integrated Elasticsearch for full-text search, including authenticated HTTPS communication, mounted TLS certificates, CA verification and reindexing of application data when search indexes are recreated.
- Integrated LibreTranslate and Flask-Babel for multilingual post translation and internationalisation, with LibreTranslate running as an independently containerised service within the Compose network.
- Authored and validated an OpenAPI 3.0 specification covering authentication, request bodies, pagination, reusable schemas, error responses and API contracts. Rendered the documentation with Scalar and published it through GitHub Pages.

- Deployed the Flask application and MySQL database to Railway using production environment variables and Gunicorn; retained Elasticsearch and LibreTranslate as local/self-hosted services due their higher storage and memory requirements.

NATS.io based Distributed Messaging System (Java Microservices Project)

While working as a Java Developer I led this project from scratch, starting with no prior experience in NATS.io or messaging systems. I independently researched and learned NATS.io, explored its pub-sub based request-reply patterns, streams, storage and so on, and translated that understanding into a working distributed system messaging platform.

I led this project as part of my work in distributed systems, taking responsibility for the overall architecture, message design, and system integration strategy. I worked closely with peers who developed individual services, aligning everyone around a shared communication pattern and ensuring that the NATS.io pub-sub based request–reply flow worked reliably across components. My role focused on coordinating the system design and making sure all parts integrated smoothly under concurrent load.

- Designed and implemented a distributed microservices communication system using NATS.io to enable fast and decoupled communication.
- Built and structured NATS.io workflows in Java, focusing on simple, scalable service interaction.
- Handled concurrent request processing using pub based listeners and multi-threaded responders to improve responsiveness under load.
- Implemented queue groups to balance requests across multiple service instances, improving reliability and preventing bottlenecks.
- Defined clear message based on JSON formats so independently built services could communicate without tight coupling.
- Tested the system under simultaneous requests to observe performance and ensure stable behaviour in realtime scenarios.
- Applied core distributed systems ideas like asynchronous messaging, event-driven design, and service decoupling in a practical setup.

MSc Artificial Intelligence Dissertation - “An AI-Driven Framework for Automated 3D Model Generation from 2D Images” **—** GitHub

Deep Learning | Computer Vision | 3D Reconstruction

As part of my MSc Artificial Intelligence dissertation, I independently designed and implemented an end-to-end pipeline for reconstructing 3D models from a single RGB image. The project combined deep learning-based depth estimation with classical geometric processing techniques, covering the complete workflow from data preprocessing to 3D mesh generation.

- Designed and implemented an end-to-end image-to-3D reconstruction pipeline using Python and PyTorch.
- Fine tuned a ResNet-50 encoder-decoder architecture for monocular depth estimation using the NYU Depth V2 dataset.
- Developed custom data preprocessing and loading pipelines, including image resizing, normalisation, batching, and dataset management for efficient model training.
- Trained and evaluated the depth estimation model, monitoring performance through quantitative metrics and qualitative visualisation of predicted depth maps.

- Generated dense point clouds from predicted depth maps by combining depth information with the corresponding RGB images.

- Reconstructed 3D meshes from generated point clouds using Poisson Surface Reconstruction**,** followed by normal estimation and mesh refinement to improve reconstruction quality.

- Built a modular pipeline separating data preprocessing, model training, inference, point cloud generation, and mesh reconstruction, improving maintainability and reproducibility.

- Resesarched the integration of deep learning and geometric processing techniques for generating meaningful 3D representations from monocular images.

BSc Computer Networking and IT Security – Final Year Project: "Sign Language to Speech Glove" **—** GitHub
IOT | Machine Learning | Assistive Technologies

This project explored how wearable sensing and machine learning could be used to improve communication accessibility for individuals who use American Sign Language (ASL). The aim was to build a system that could recognise hand gestures and translate them into speech in real time.

- Designed a wearable glove based system using flex sensors and an MPU6050 gyroscope and microcontroller to capture finger movements and hand orientation.

- Created and organised a custom dataset of ASL alphabet gestures, using 11 sensor readings as input features.

- Treated gesture recognition as a supervised learning problem and implemented a K-Nearest Neighbours (KNN) model for classification.

- Built a complete pipeline from data collection and preprocessing to model training and testing and predciting.

- Developed a real-time system that converts predicted gestures into text and then into speech using a textto-speech API (Voicerss).

- Integrated hardware and software components into a working end-to-end system for live gesture-tospeech translation.

- Focused on making the system usable in real-world scenarios to support communication between sign language users and non users.

- This project helped me understand how machine learning can be applied in practical assistive technologies and human centred systems.

Animal Image Classification System — GitHub **Deep Learning | Computer Vision**

This project involved building a deep learning model to classify animal images into different categories. The main goal was to understand how neural networks can learn patterns from images and use them for accurate classification.

- Built an image classification pipeline to sort animal images into predefined classes.

- Used a Kaggle dataset and applied basic preprocessing such as resizing, normalisation, and data augmentation.

- Designed and trained a convolutional neural network (CNN) to learn visual features from images.
- Implemented the full workflow from data preparation through to training and evaluation.
- Tested and evaluated model performance using standard accuracy based metrics.
- Experimented with different training settings to improve model results.
- Gained practical experience in applying deep learning to real-world image recognition tasks.

Experience

Assistant Java Developer FocusOne Payment Solutions (Fintech) 07/11/2022 – 31/10/2023
Contributed to backend systems powering real-time credit and debit card authorisation for a fintech platform.

- Designed and maintained high-performance REST APIs using Java and Spring Boot for a platform processing real-time card authorisations across thousands of daily transactions.
- Refactored legacy applications through root-cause debugging and targeted code improvements, reducing production incidents and improving system reliability.
- Evaluated NATS messaging for event-driven architecture adoption, presenting findings that shaped the team's infrastructure roadmap for low-latency, fault-tolerant systems.
- Built and maintained Swagger/OpenAPI documentation and contributed to testing of API endpoints (Postman), cutting onboarding time and improving cross-team collaboration on API contracts.
- Delivered new backend features aligned with evolving product requirements in a fast-paced Agile environment, participating in code reviews and sprint planning.
- Authored system specifications and API design documentation supporting long-term maintainability, reliability, and regulatory compliance.

3D Animator Hive Animation Studios Pvt Ltd, Nepal June 2016 – December 2020
Worked across the full 3D production pipeline including modelling, texturing, animation, lighting and rendering

- Developed strong understanding of 3D geometry, spatial relationships, and scene composition.
- Used softwares such as Autodesk Maya, 3Ds Max, zBrush to create and refine 3D assets and animations.
- Applied lighting, shading, and rendering techniques to produce realistic visual outputs.
- Gained professional experience in animation principles.
- Built strong spatial thinking and structured visual problem-solving skills.

Teacher Mathematics Teacher (Primary Level) Corban Boarding School, Nepal 2013 – 2014

- Taught basic mathematics to Grade 3 and Grade 4 students.
- Supported students in developing foundational arithmetic and problem solving skills.

- Prepared and delivered simple, structured lessons tailored to young learners.
- Developed strong communication, patience, and classroom management skills.

Volunteering Experience

Community 3D Visualization Project – 2018 – youtube
In collaboration with local ward authorities (Budiganga Gaupalika) in Nepal.

- Volunteered to create a 3D visual concept of a proposed community development plan, including a local ground, old age care home, children's playground, and health post.
- Translated urban planning ideas into a visual model to help support discussions and decision making at community level.
- Worked independently to create a clear and structured 3D design as per the requirements and a walk through animated video.
- Used my 3D modelling skills to contribute to a local civic initiative focused on improving public infrastructure planning and community engagement.
- Applied spatial thinking and visual design skills to a real world problem with social impact.

Digital Literacy and Internet Safety Tutor
Janpriya Public School, Nepal - 2023

- Delivered a short training programme for students in Grade 8 to Grade 10 focused on internet safety, digital awareness, and responsible use of technology.
- Introduced basic ideas of information systems and simple programming concepts in an easy to understand way.
- Taught students about online privacy, safe browsing habits, and managing their digital footprint.
- Designed and delivered sessions in a way that was engaging and suitable for secondary school students.
- Improved my ability to explain technical ideas in a simple and clear manner and communicate effectively with younger learners.

Skills

I have experience across software development, intelligent systems, and 3D visual computing, gained through academic work, industry roles, and independent projects.

Professional

- **Languages:** Python, Java, JavaScript, MATLAB
- **Testing & QA:** Unit testing, integration testing, pytest (familiar), test case design
- **CI/CD & DevOps:** Git, GitHub, CI/CD pipeline concepts, Docker (foundational), Linux
- **Backend & APIs:** Spring Boot, Django, REST API,, Java Spark, NATS (event-driven messaging), Swagger/OpenAPI

- **Frontend:** Vue.js, HTML, CSS, JavaScript
- **Machine Learning / AI:** PyTorch, TensorFlow, Keras, Scikit-learn, OpenCV, Open3D
- **Databases:** MySQL, SQLite, PostgreSQL
- **Concepts:** Distributed Systems, API Design, Agile/Scrum, Network Security, Computer Vision

Research

- **Methodology Design:** Experience designing and implementing research methodologies, including problem formulation, literature informed system design, experimental planning, model development, and performance evaluation through academic research projects.
- **Critical Thinking:** Ability to critically evaluate existing approaches, identify limitations, compare alternative methods, and make informed design decisions based on evidence and experimental findings.
- **Data Analysis and Interpretation:** Experience analysing experimental results using quantitative performance metrics and qualitative evaluation to assess model behaviour, identify trends, and guide iterative improvements.
- **Information Literacy:** Strong ability to locate, evaluate, and synthesise information from academic literature, technical documentation, and research publications to support independent learning and evidence based decision making.
- **Research Communication:** Experience documenting research methodology, presenting technical concepts clearly, and communicating findings through dissertations, project reports, and technical documentation for both technical and non technical audiences.
- **Independent Research:** Demonstrated ability to independently learn unfamiliar concepts, investigate new technologies, and apply research findings to solve complex technical problems.
- **Project and Time Management:** Experience managing multiple academic and professional commitments while planning research activities, meeting project milestones, and delivering work within established deadlines.

Other

- Communication Skills
- Critical Thinking
- Analytical Thinking
- Problem Solving
- Independent Learning
- Time Management
- Adaptability
- Attention to Detail
- Team Collaboration

- Languages: Nepali(Native), English(Advanced), Hindi(Advanced)

Certifications & Professional Development

- Google Data Analytics Specialisation — Coursera (2023)
- API Designer & API Security Architect — API Academy
- AWS Academy Cloud Foundations — Amazon Web Services
- Deep Learning Onramp & Machine Learning Onramp — MathWorks
- Data Science Foundations — LinkedIn Learning

References

1. Dr. Shaheen Khatoon (Senior Lecturer)
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